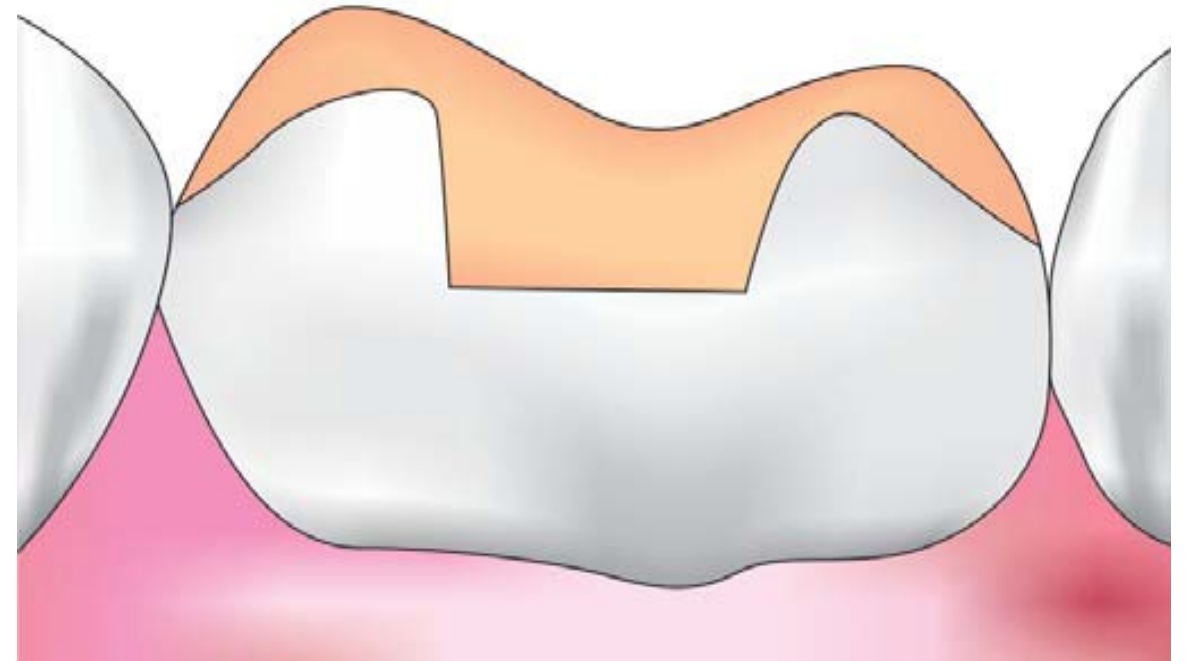




Outline:

- ❖ Introduction
- ❖ Indications Of Metal Inlay And Onlay
- ❖ Contraindications Of Metal Inlay And Onlay
- ❖ Advantages Of Cast Metal Restorations
- ❖ Disadvantages Of Cast Metal Restorations
- ❖ Principles Of Tooth Preparation For Cast Metal Inlay
- ❖ Steps Of Tooth Preparation For Inlay
- ❖ Technique For Making Cast Metal Restoration
- ❖ Casting Defects

Introduction

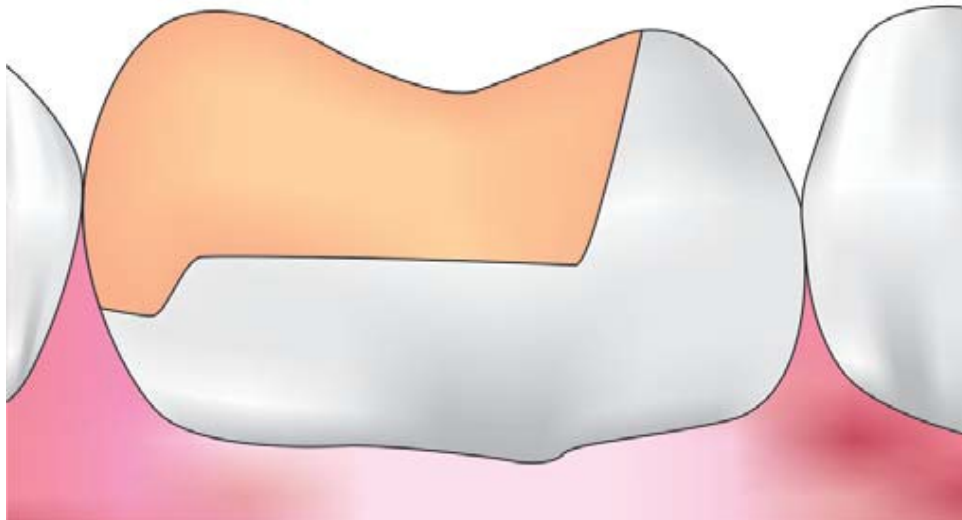
Cast metal restorations:

- Introduced in dentistry about 100 years ago and soon they became a subject of great interest to both dentists and patients.
- It became most popular from the 1930s to the 1970s because of their acceptable functions and restorative longevity.
- Are made from alloys formed by combination of gold with other metals such as *silver, copper, zinc, platinum and palladium*.
- They are mainly used in three forms:
 1. Inlay
 2. Onlay
 3. Partial crown.

DEFINITION

1. Inlay:

- An inlay is an **indirect intracoronal restoration** which is fabricated extraorally and cemented in the prepared tooth.
- Inlay may cover **none, or all** but one cusp of a tooth.

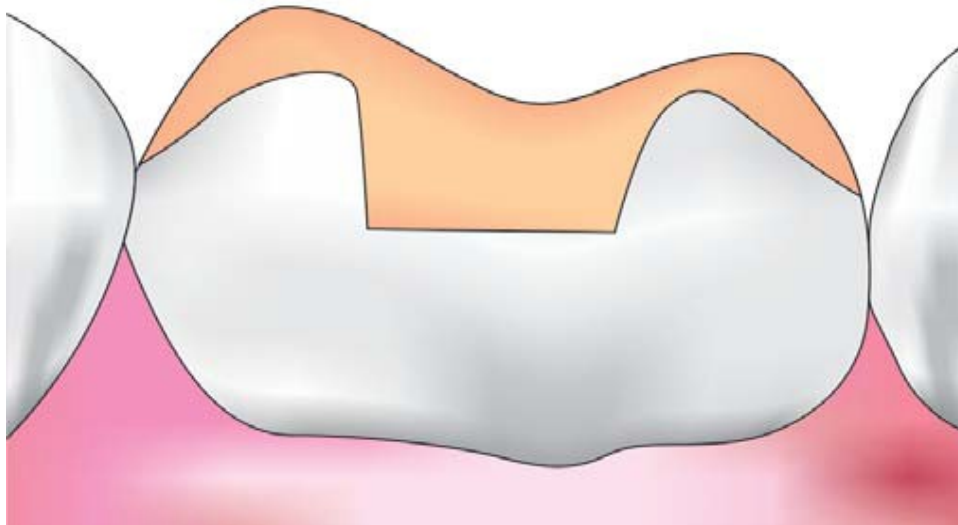


Class II inlay

Con't.....

2. Onlay:

- An onlay is a combination of **intracoronal and extracoronal** cast restoration which covers one or more Cusps.



Onlay

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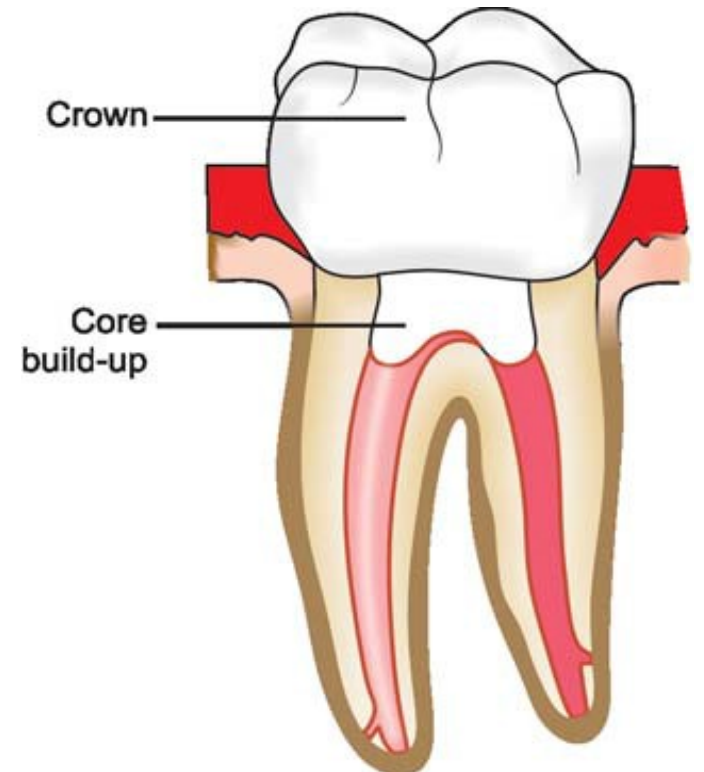
3. Partial crown:

- In partial crown, a part of the crown remains uncovered and rest of the crown is covered
like three quarter crown and seven-eighth crown.

4. Crown:

- A crown completely covers the crown of the tooth

Crown given on root canal treated tooth



INDICATIONS OF METAL INLAY AND ONLAY

- **In extensive proximal surface caries** in posterior teeth involving buccal and lingual line angles.
- In patients with **good oral hygiene and low caries index**.
- **Post endodontic restoration** are preferably restored by **onlay** to strengthen the **remaining tooth structure** and to distribute occlusal forces.
- **When fracture line is present in enamel and dentin.**
Inlay/onlay can brace the tooth and prevent fragmentation of the tooth.
- To maintain and restore **proper interproximal contact and contour and for occlusal plane correction.**

Cont.....

- **Abutment teeth** of removable partial denture are indicated for onlay because they provide superior physical properties to withstand the forces imparted by the partial denture.
- In posterior teeth with **heavy occlusal forces and attrition.**

CONTRAINDICATIONS:

- Where **esthetics is prime consideration** because metal inlay and onlay display metal color.
- Patient having high caries index.
- When patient cannot come for second visit.
- In young patients, usually direct restorations are preferred since they require longer and more number of appointments. Also the **chances of iatrogenic pulp exposure are more in these patients** because of high pulp horns.
- **Where expected life of a tooth is short**, i.e. periodontally involved teeth in aged persons.

Cont.....

- **For patient of low economic status**, inlay and onlay are not given because of higher costs.
- In patients having **restorations with different metals** since dissimilar metals cause galvanic currents when they come in contact with each other.

ADVANTAGES

- ❑ **Better reproduction of contacts and contours**, since they are fabricated by indirect technique,
- ❑ **Much more wear resistant than direct composite restorations**, especially when restoring the occlusal surface.
- ❑ **More biocompatible with better tissue response.**
- ❑ **Strengthens remaining tooth structure.**
- ❑ **Less chairside time is required.**
- ❑ **Build in bulk and not in increments**, there are less chances of voids and internal stresses.
- ❑ **Extraoral polishing is easy.**



DISADVANTAGES

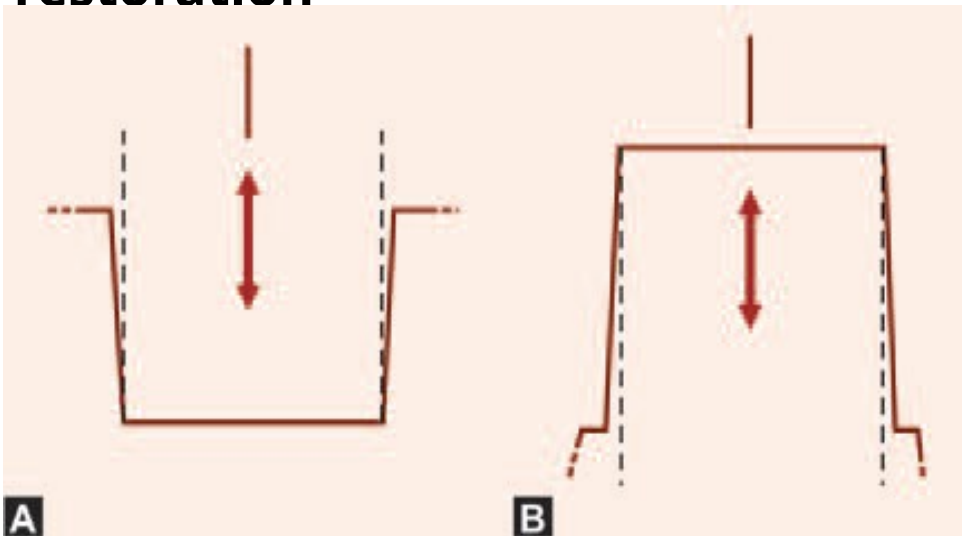
- ❑ It requires patient's subsequent appointment, with need for temporary restoration.
- ❑ More expensive than direct restorations.
- ❑ They are more technique sensitive.
- ❑ Repair is difficult with these restoration.
- ❑ Bonding of these restoration to tooth is weak at tooth cement casting junction. It can lead to microleakage of the restoration with time.
- ❑ Cast metal restorations are not acceptable esthetically.

PRINCIPLES OF TOOTH PREPARATION FOR CAST METAL INLAY

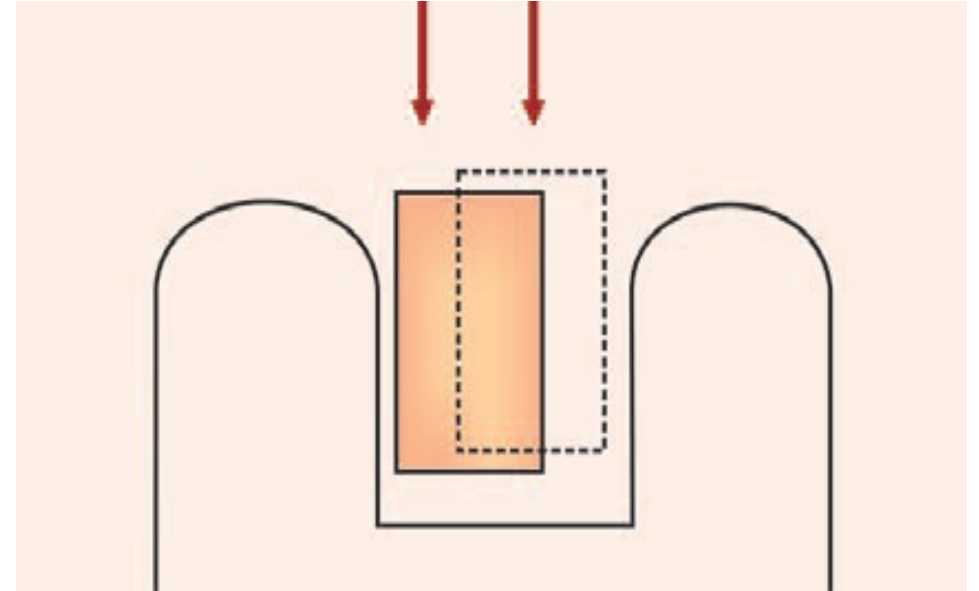
- **Preparation Path**
- **Inlay Taper**
- **Circumferential Tie**

Preparation Path

- Preparation should have **single insertion path opposite to the occlusal load and parallel to the long axis of tooth** . This helps in retention of the restoration



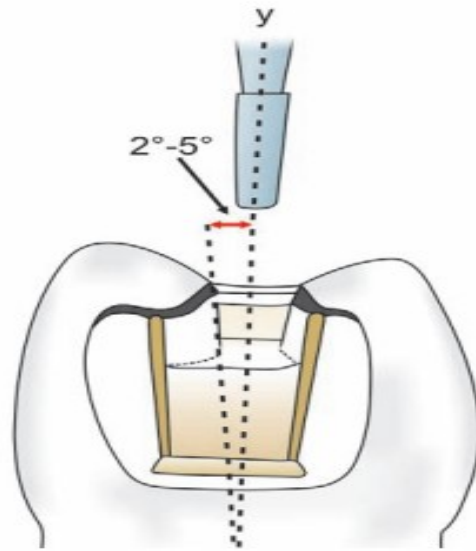
Inlay preparation should have single path of insertion parallel to long axis of tooth



With multiple paths of insertion, there is no retention for inlay even if walls are parallel to each other

Inlay Taper

- To have **unhindered removal and placement** of the wax pattern and seating of the final casting, intracoronal and extracoronal tooth preparation should have some taper.
- Ideally, a tooth preparation should have **slight diverging walls from gingival to occlusal surface**.
- **Tape**

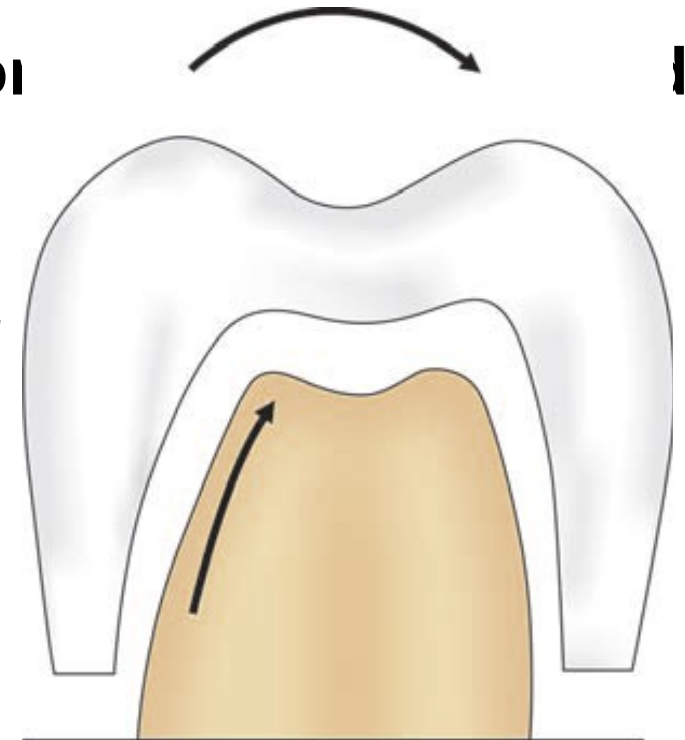


Parallelism of opposing walls and slight occlusal divergence provides retention of inlay

A 2°-5° of taper provides optimal retention for inlay

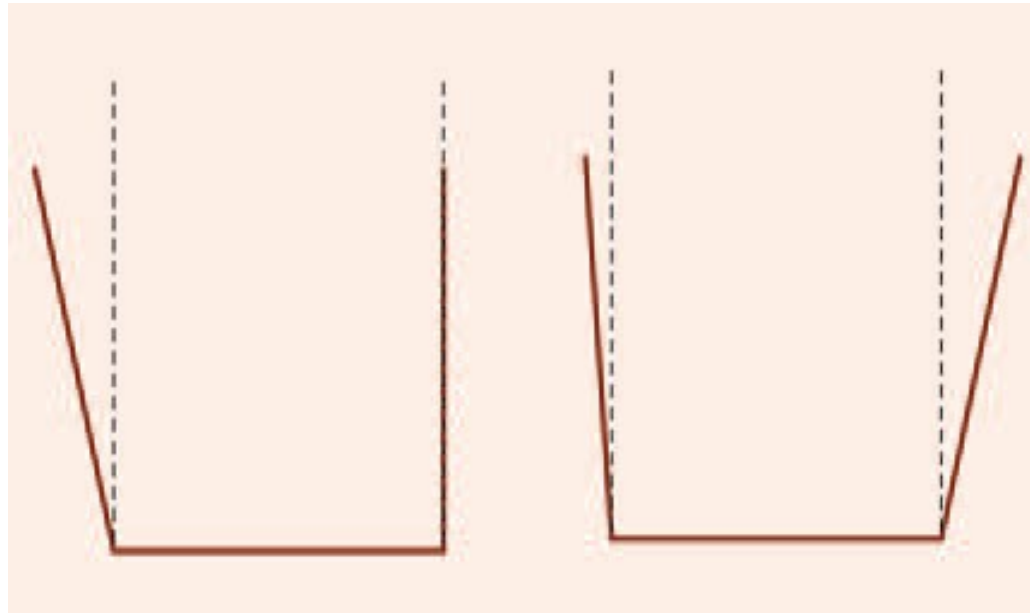
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- **The optimal taper** should lie in the range of **2 to 5° per wall**. But
 - If **longitudinal** walls of preparation are short, **a maximum of 2° taper is given**, and
 - If longitudinal height of preparation is more, it **also be increased**,
although **it should never exceed 10°**.
Giving more than required taper in preparation walls results in reduced retention of inlay



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- Also the preparation **should never have one side with more taper than the other** since it can result in *more than one path of insertion*



Uneven taper of preparation walls result in failure of inlay

Circumferential Tie

- It refers to the design of **cavosurface margin of an inlay tooth preparation.**

This junction between tooth - cement inlay is the weakest part of the cast metal restoration.

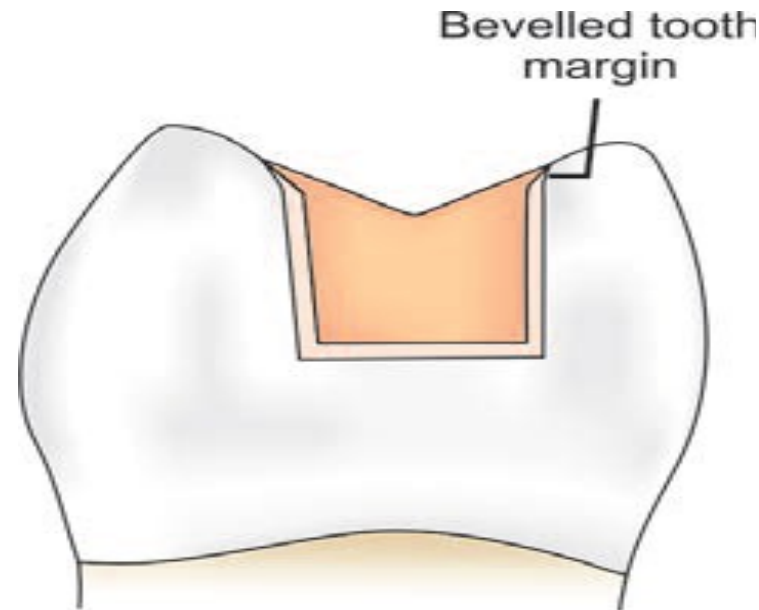
- For the success of restoration, **the margins of restoration should be designed so as to achieve its maximum adaptation to tooth structure.**
- Cavo surface margins of an inlay preparation can be of two types:
 1. Bevels
 2. Flares.

1. Bevels

Bevel : is defined as the **inclination that one surface makes with another when not at right angles.**

- Objective of bevel is **to form a metal wedge of 30 to 35°**, thus enhancing the chance to achieve closure at the interface of cast gold and tooth
- By beveling, **a strong enamel margin with an angle of 140° to 150° can be produced.**

Bevel helps in better adaptation of restoration to the tooth surface



Types of bevel

- According to shape and type of tissue involved, bevels can be:

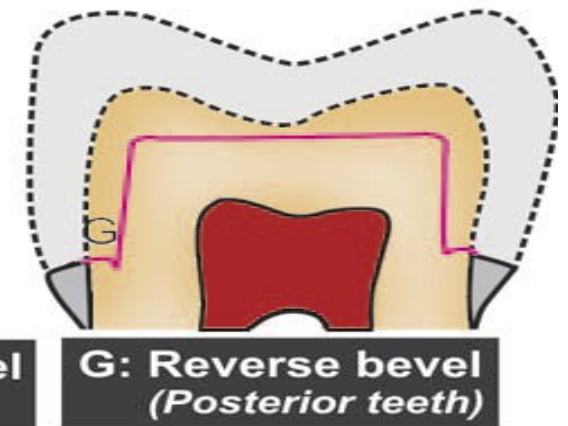
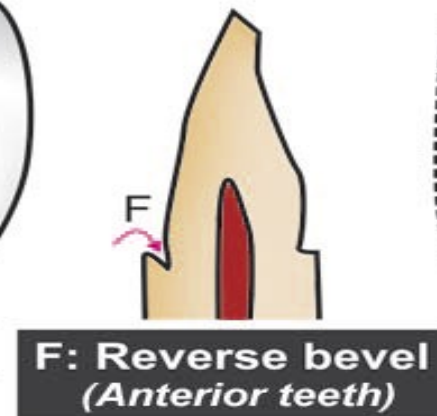
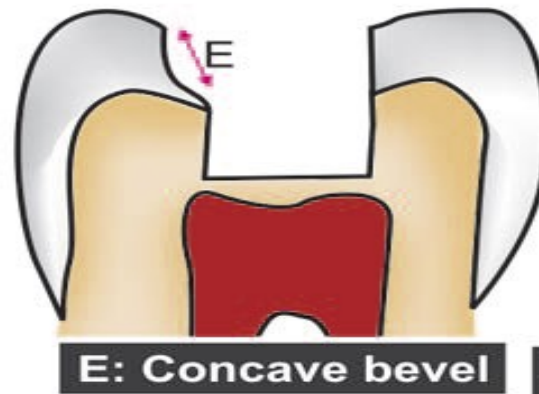
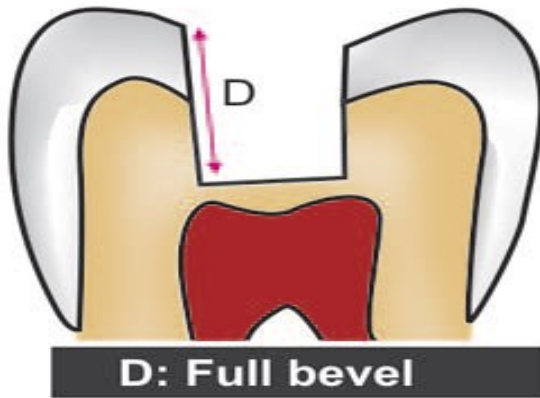
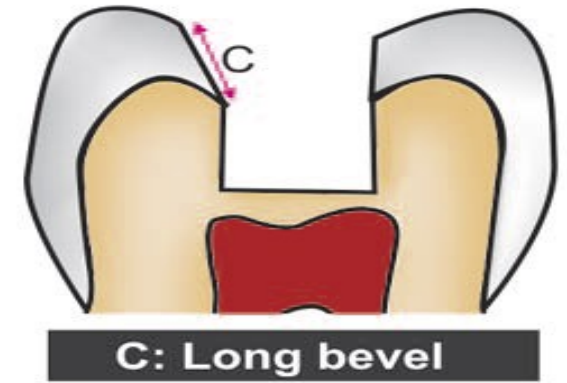
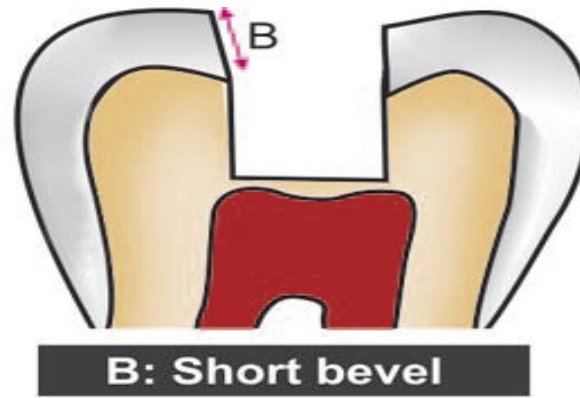
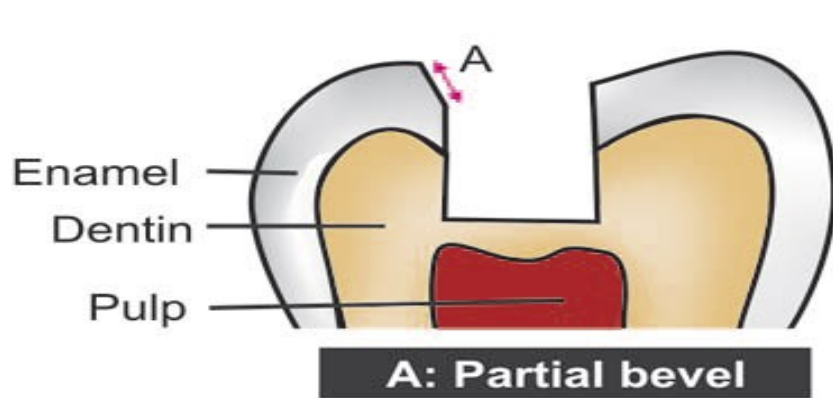
a. Ultrashort or partial bevel

- Beveling of less than two-third of the total enamel thickness.
- Used to trim the enamel rods from preparation margins.
- Used in type I casting alloys.

b. Short bevel

- Beveling of full thickness of enamel wall but not dentin.
- Used mostly for restorations with type I and II casting alloys.

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c. Long bevel

- Includes full thickness of enamel and half or less than half thickness of dentin.
- Preserves the internal 'boxed up' resistance and retention features of the preparations.
- Used in types I, II and III of cast gold alloys.

d. Full bevel

- Includes full enamel and dentinal wall.
- It deprives the preparation of its internal resistance
- Full bevel should be avoided except in cases where it is a must.

e. Hollow ground (concave) bevel

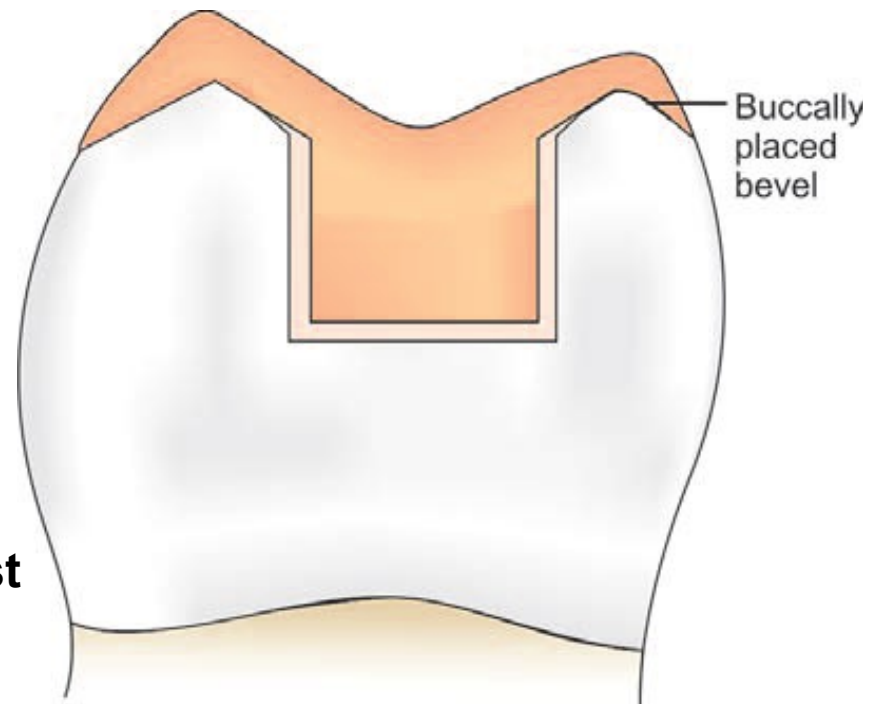
- it is concave in shape and not a bevel in true sense.
- It is rarely used.

f. Reverse or inverted bevel

- It is beveling in the reverse or inverted shape given on the gingival seat in the axial wall toward the root in anterior teeth.

Functions of bevels:

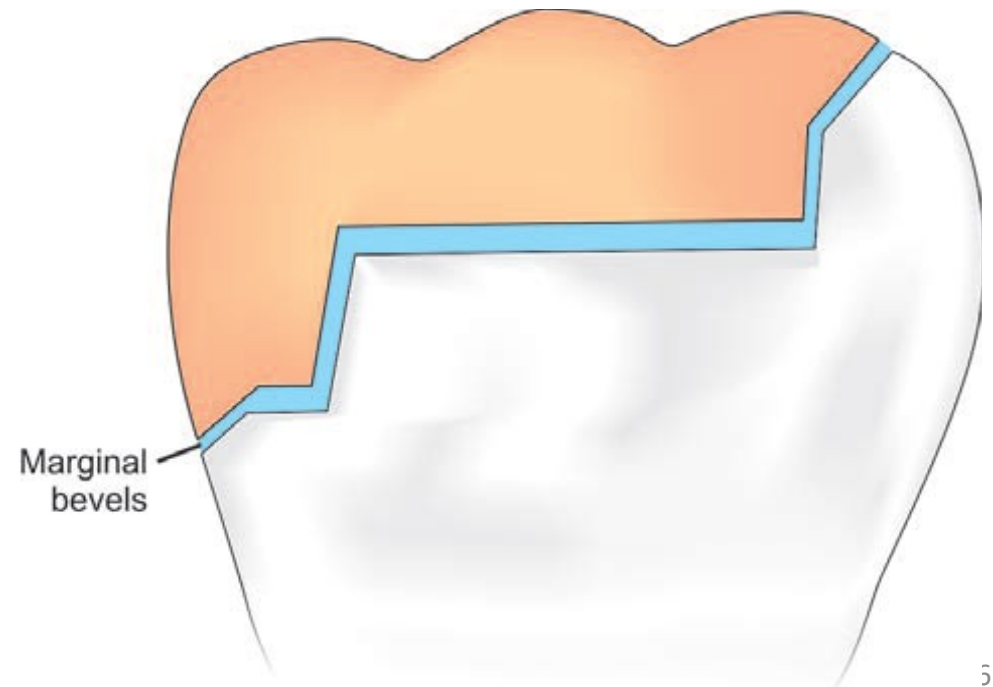
- By beveling, weak enamel is removed.
- Acute angled metal margins (35°-45°) allow the metal margins to be burnished against tooth surface



Bevels allow better finishing and burnishing of metal margins against tooth

- **Beveling increases retention, resistance, aesthetics and color matching.**
- **It improves junctional relationship between the restorative material and tooth**
- **The types of margin of tooth preparation depends mainly on the compressive strength, edge strength and tensile strength of the restorative material.**

Marginal bevels reduce the space between tooth structure and the restoration, thus help in better retention



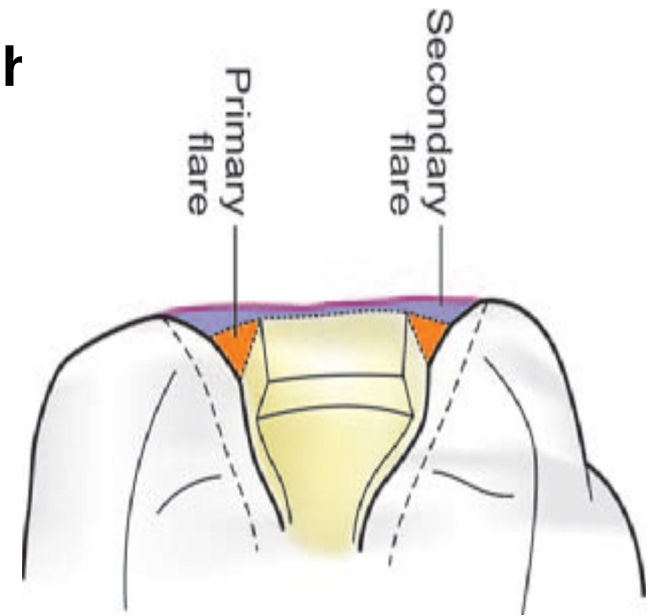
2. Flares

- **Flares:** are concave or flat peripheral portions of the facial or lingual proximal walls.
- These are of two types:

A. Primary flare:

- It is basic part of circumferential tie.
- It is like a long bevel and is always directed 45° to the proper

Primary flare and secondary flare



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Indications:

- Primary flare is indicated when **normal contacts are present**.
- When there is **minimal extension of caries in buccolingual direction**.

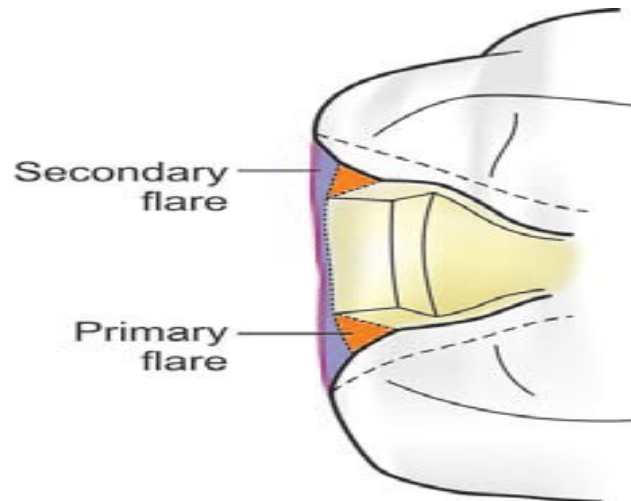
Functions:

- Weak enamel is removed.
- **Improves junctional relationship** between the restorative material and tooth.
- Maintains the **marginal seal**.
- Brings the **facial and lingual margins to cleansable and finishable areas**.

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b. Secondary flare:

- It is a **flat plane superimposed peripherally to the primary flare**
- **Secondary flare is not given in the areas where aesthetics is more important.**



Primary flare and secondary flare

Indications of secondary flares:

- When **broad contact area** is present.
- To include the **faults present on facial and lingual walls beyond primary flare**
- When caries are **widely extended in buccolingual direction**.
- To include the **undercuts present at cervical aspect of facial and lingual walls**.

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Advantages of secondary flare:

- It encourages **self-cleansing margins** because it is extended into the embrasures.
- Permits **easy finishing** of the restoration.
- Allows **better burnishing** of the metal.
- Produces **more blunted and stronger margins**.

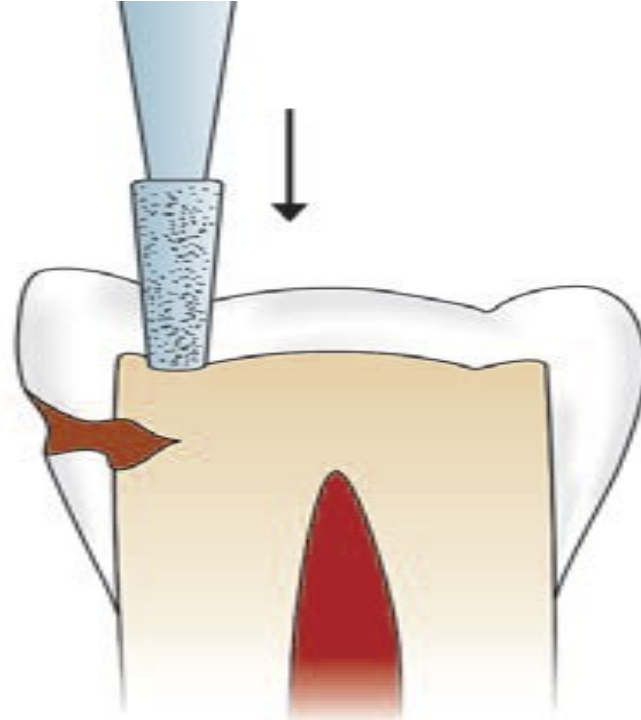
STEPS OF TOOTH PREPARATION FOR INLAY

- **Occlusal Outline Form**
- **Proximal Box Preparation**
- **Resistance and Retention Form**
- **Placement of Grooves, Bevels and Flares**

Occlusal Outline Form

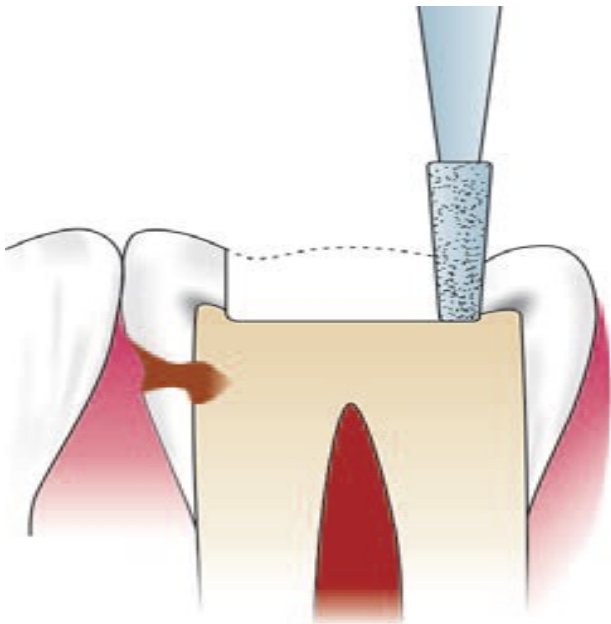
- The entry point: should be **closest to the involved marginal ridge** (for example, mesial)
- Keeping the bur parallel, extend the tooth preparation while maintaining the **initial pulpal depth of 1.5 mm.**

Penetrate the bur closest to the involved marginal ridge

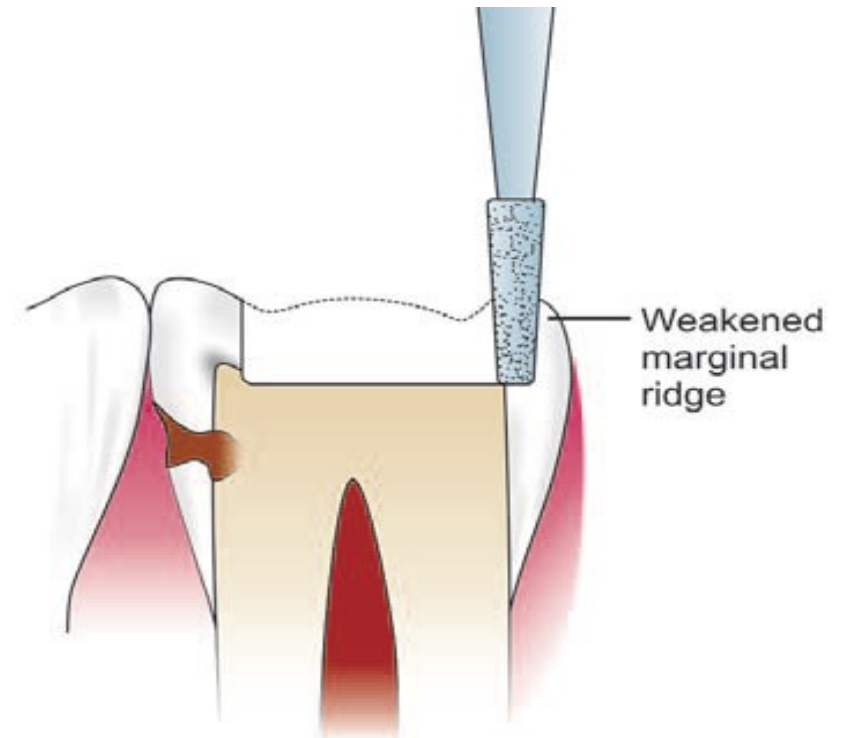


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- **make a flat pulpal floor of proper depth and the occlusal walls of a uniform taper** are being prepared with the same bur
- Care is used to avoid overcutting



Keeping the same depth (1.5 mm),
establish the occlusal outline

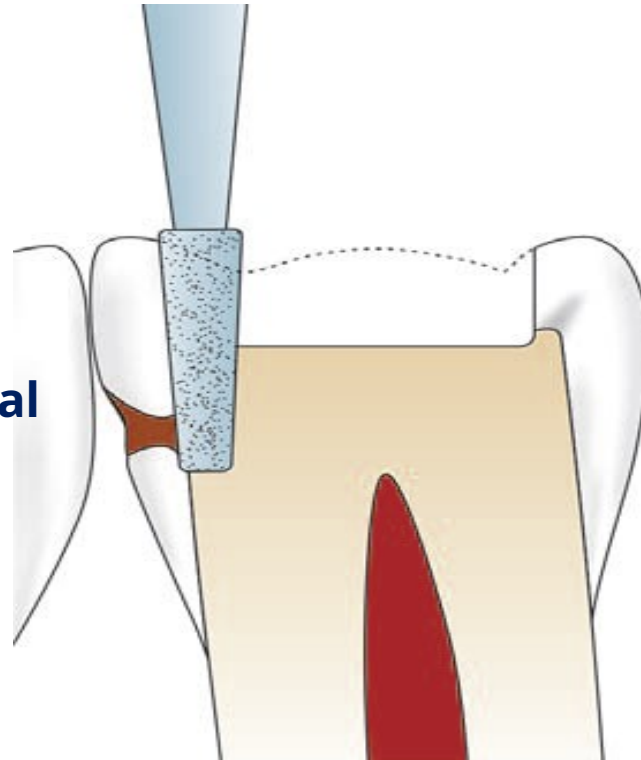


Overcutting can result in
weakening of marginal ridge

Proximal Box Preparation

- Using the same bur, isolate the **distal enamel by proximal ditch cut**. Width of this cut should be **0.8 mm with 0.5 mm in dentin and 0.3 mm in enamel**.
- Extend this ditch facially and lingually to the sound tooth structure and proceed gingivally

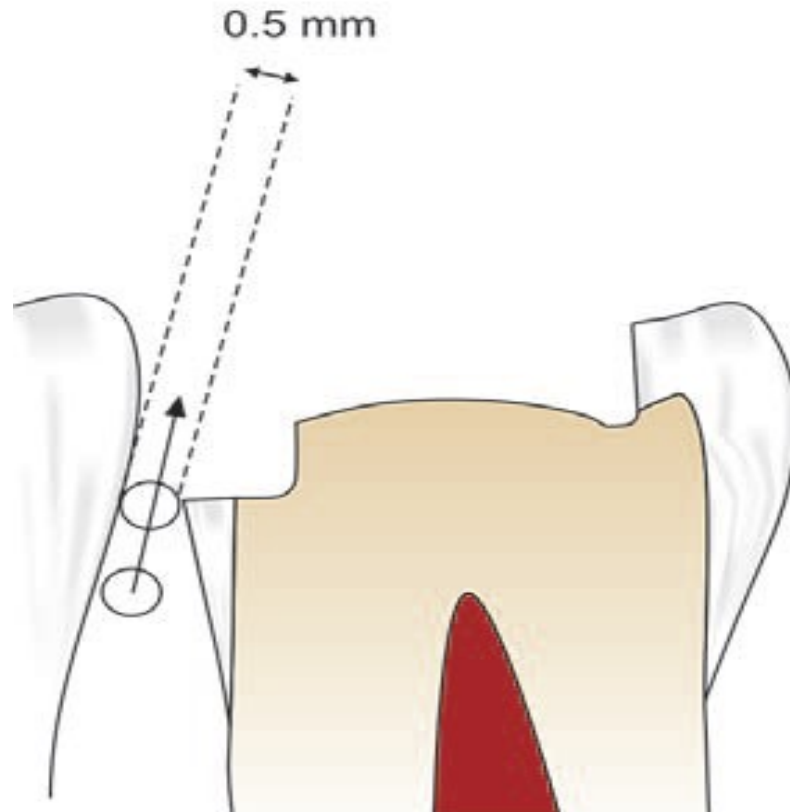
The proximal ditch is given after occlusal preparation



Con't.....

- **Gingival extension:-** should remove any caries present on the gingival floor and

tooth
It should provide at least 0.5 mm clearance from the adjacent



The gingival floor should provide at least 0.5 mm clearance from the adjacent tooth

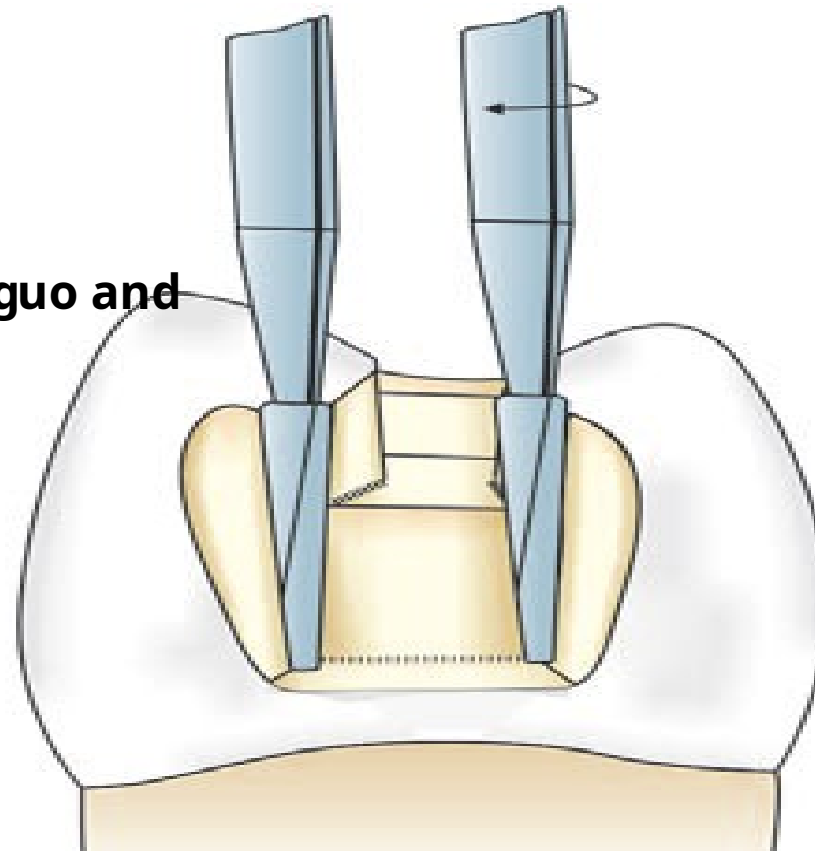
Resistance and Retention Form

- A nicely designed inlay preparation should have **proximal box form** for retention and **flat pulpal and gingival walls** for resistance to dislodging forces.
- The extra thickness of gold provided by the box form and the occlusal step of the preparation helps in increasing its retention and resistance.
- The occlusal dovetail of the preparation also provides added retention to the restoration.
- The resistance and retention form of the occlusal step of the preparation will often require variations in design depending upon a careful study of the size and position of the pulp horns.

Placement of Grooves, Bevels and Flares

- Retention grooves if needed are placed in the facioaxial and linguoaxial line angles

Retention grooves are placed in linguo and facio axial line angles



ADDITIONAL RETENTION AND RESISTANCE FORM FEATURES FOR CAST RESTORATION

1. Grooves
2. Reverse bevel
3. Internal box
4. External box
5. Pins:
6. Slot

TECHNIQUE FOR MAKING CAST METAL RESTORATION

PROCEDURES

- **Impression Taking for Cast Metal Restoration**
- **Working Cast**
- **Record of Interocclusal Relationship**
- **Working Die**
- **Temporary (Interim) Restoration**
- **Wax Pattern Fabrication**

- **Investing**
- **Washing of Wax Pattern**
- **Casting**
- **Procedure**
- **Cleaning of Casting**
- **Trying in the Casting**
- **Cementation of the**

CASTING DEFECTS

- **The various steps in making of casting should be followed systematically, otherwise chances of casting defects are increased.**
- **Casting defects are of many types and may be classified as:**
 - **Surface Roughness and Irregularities**
 - **Distortion**
 - **Incomplete Casting**
 - **Discoloration**
 - **Porosity**

Surface Roughness and Irregularities

- **Surface roughness is generalized roughness of the surface of the casting while surface irregularities are isolated imperfections such as nodules, ridges, fins and spines.**

Causes

- Inadequate water/powder ratio.
- Too rapid heating of investment.
- Prolonged heating of investment.
- incomplete elimination of wax.
- Air bubbles on the pattern during investment.
- Direction of sprue former.
- Placement of several patterns in a ring together.
- High temperature of molten alloy.
- High casting pressure.
- Presence of foreign bodies.

Prevention

- **Powder and water ratio should be adequate.**
- **Heating should be done gradually as too rapid heating would cause flaking of the investment material.**
- **Gypsum bonded investment should not be heated above 700°C as it would cause disintegration of the investment material.**
- **Ring should be heated for adequate period of time for complete elimination of wax.**
- **Air bubbles can be avoided by vibrating mixture before and after mixing.**

Cont.....

- **The direction of sprue former should be adjusted to 45°.**
- **Several pattern should not be placed closely together in a single ring as expansion of wax can cause breakdown of investment material.**
- **High temperature of molten alloy causes surface roughness.**
- **Foreign bodies should be avoided during investment of material.**

Distortion

- Distortion of casting usually occurs due to distortion of inlay wax pattern. The coefficient of thermal expansion of inlay wax is high.
- Distortion of wax pattern increases if it is improperly handled. Time lag between making of pattern and investing is also crucial during casting.

Causes

- Distortion of wax pattern.
- Mishandling of pattern.
- Due to unequal movement of walls of wax pattern while the investment is setting.

Prevention

- Proper manipulation of wax pattern

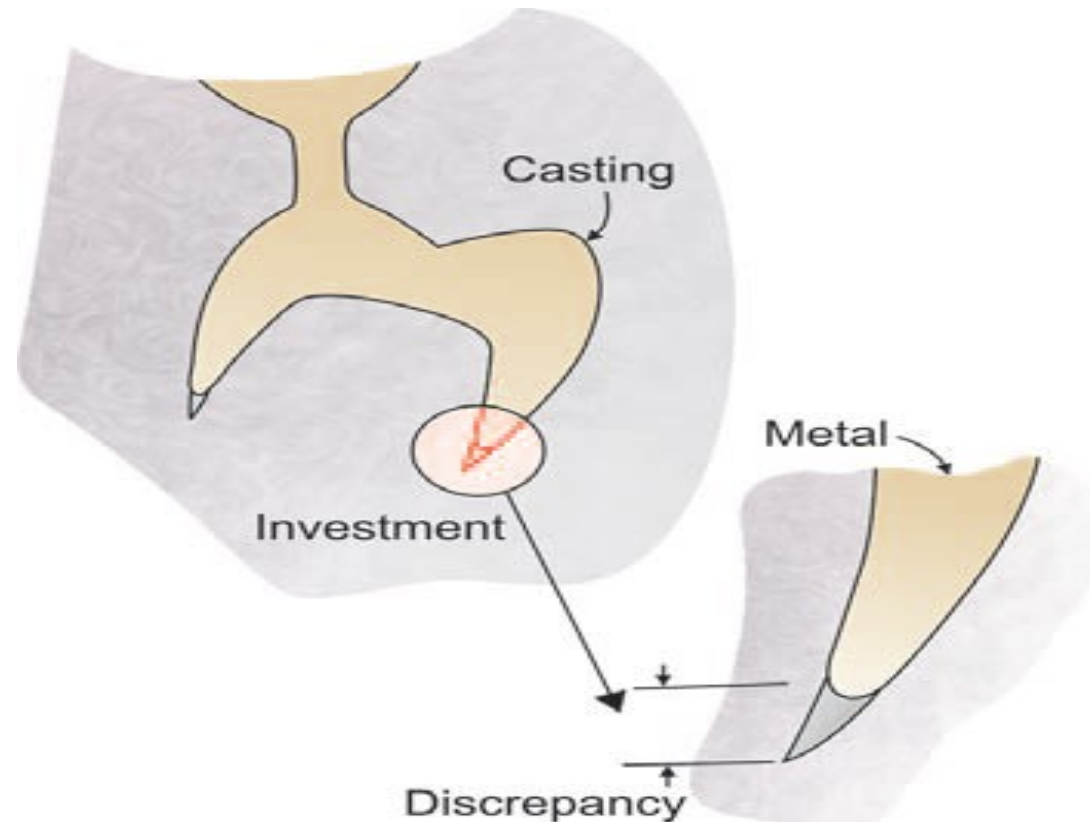
Incomplete Casting and Rounded Margins

Causes

- Inadequate heating of the alloy.
- Inadequate casting pressure.
- Improper length and diameter of sprue.
- Improper burnout of wax pattern.
- Insufficient molten alloy.

Prevention

1. Alloy should be heated above fusion temperature, i.e. 570°C.
2. Casting pressure should be adequate to force the molten alloy.
3. Proper length and diameter of sprue former.
4. Proper burnout of wax pattern should be done so that no carbon residues left in the mold.



Incomplete casting

Discoloration

- The casting usually appears dark after removing from the investment due to presence of oxides. This can be usually be removed by process known as 'Pickling'.

Causes

- *Prolonged heating:*
- Sulfur content of torch flame also affects the casting.
- Under heating of the investment also leaves the wax residues in the casting, affects the color of the casting.
- Contamination with copper during the process of pickling causes discoloration.

Prevention

- Avoid prolonged heating of the investment.
- Change the source of flame.
- Proper heating should be done.
- Tips of tongs must be covered with rubber to

Porosity

- Porosity is considered as a major defect in the casting which can occur on the internal as well as on the external surface of the casting. It usually weakens the casting.
- Various types of porosities can be:

TYPES			
• Solidification shrinkage porosity a. Localized shrinkage porosity b. Microporosity	• Gaseous defects a. Gas inclusion porosity b. Pinhole porosity	• Backpressure porosity	• Subsurface porosity.

PIN RETAINED CAST RESTORATIONS

- **Pins are generally used for providing additional retention.**
- **Cast pin channels are wider and have slightly divergent walls in comparison to the pin holes prepared for pin retained amalgam restorations.**
- **The cast pin channels are prepared with the help of tapering fissure bur having a diameter of 1 mm with the depth of about 2 to 3 mm.**

Indications for Pin Retained Cast Restorations

Pin retained cast restorations are indicated in following cases:

- **When occlusogingival height is very short.**
- **In cases of excessively tapered tooth preparation.**
- **Cuspal fractures where large occlusal inlays and onlays are to be prepared.**
- **When the proximal box is very long. The pin channel is prepared at the other end of occlusal lock.**
- **In full crown preparation when one wall is very short and another wall is very long.**
- **For shallow and wide preparations, when it is not possible to place surface extensions for retention.**



